What is a Hacker

Hackers usually have strong knowledge of operating systems, networking, programming, security vulnerabilities, and web technologies. This technical background helps them understand how systems work and how weaknesses can be discovered.

Types of Hackers

Hackers are generally classified into three main categories.

White Hat Hacker (Ethical Hacker)

White hat hackers are security professionals who use hacking techniques legally to identify vulnerabilities in systems. Their goal is to improve security and protect organizations from cyber-attacks.

They often work as penetration testers, security researchers, cyber security analysts, and bug bounty hunters.

Example: Testing a company's website security with permission.

Black Hat Hacker

Black hat hackers perform illegal hacking activities. Their goal is usually to steal data, gain unauthorized access, or cause damage.

Common activities include data theft, ransomware attacks, website defacement, identity theft, and financial fraud. These hackers violate laws and regulations.

Grey Hat Hacker

Grey hat hackers fall between white hat and black hat hackers. They may access systems without permission but usually do not have malicious intentions. They might find vulnerabilities and report them to the company but without official authorization.

What is an Operating System (OS)

An Operating System is system software that manages computer hardware and software resources. It acts as an interface between the user and the computer hardware. Without an operating system, a computer cannot function properly.

The operating system performs several tasks such as managing hardware resources, running applications, managing files and storage, handling input and output devices, and managing memory and processes.

Examples of Operating Systems:

- Windows
- Linux
- macOS
- Android
- iOS

Types of Operating Systems

Operating systems can be categorized based on their usage and functionality.

Desktop Operating System

Used for personal computers and laptops. Examples: Windows, macOS, Linux.

Mobile Operating System

Designed for smartphones and tablets. Examples: Android, iOS.

Server Operating System

Used to manage servers and network resources. Examples: Linux Server, Windows Server.

Embedded Operating System

Used in devices such as routers, smart TVs, and IoT devices. Examples: Embedded Linux, RTOS.

What is Linux

Linux is an open-source operating system based on the Unix architecture. It is widely used by developers, system administrators, and cyber security professionals. Linux is free, customizable, and highly secure. Because of these features, it is widely used in cyber security and penetration testing.

Linux allows users to modify the source code and customize the system according to their needs.

Why Linux is Used in Hacking

Linux is the most commonly used operating system for cyber security professionals and ethical hackers. Reasons why Linux is preferred include:

- **Open Source:** The source code is publicly available and can be modified.
- **Powerful Command Line:** Linux provides a powerful terminal that allows users to perform complex tasks efficiently.
- **Security and Stability:** Linux systems are generally more secure and stable compared to many other operating systems.
- **Tool Availability:** Many cyber security tools are built specifically for Linux environments.
- **Customization:** Linux allows complete control over system configuration.

Linux Distributions (Distros)

A Linux distribution is a version of Linux that includes the Linux kernel along with additional software and tools. Different distributions are designed for different purposes.

Examples of Linux distributions include Kali Linux, Ubuntu, Parrot OS, Debian, Fedora, and Arch Linux. Each distribution serves a specific purpose such as development, penetration testing, or general use.

Kali Linux

Kali Linux is one of the most popular Linux distributions used for cyber security and penetration testing. It is developed and maintained by Offensive Security.

Kali Linux contains hundreds of pre-installed security tools used for penetration testing, digital forensics, security research, vulnerability assessment, and wireless attacks.

Popular tools in Kali Linux include Nmap, Burp Suite, Metasploit, Wireshark, John the Ripper, and Aircrack-ng.

Lab Setup for Kali Linux

Before practicing hacking techniques, it is important to create a safe and controlled lab environment. A cyber security lab allows students to practice attacks and defenses without harming real systems.

Common lab components include Kali Linux as the attacker machine, a vulnerable machine as the target, and virtualization software. Virtual lab environments allow safe experimentation.

Types of Kali Linux Installation

- **Virtual Machine Installation:** Safe testing environment, easy to install, and no need to modify the main operating system.
- **Dual Boot Installation:** Full hardware performance and can choose OS during boot.
- **Live Boot (USB):** Portable and does not modify the system.
- **Bare Metal Installation:** Maximum performance and full control over hardware.

Ethical Hacking and Learning Path

To become a cyber security professional, students should focus on networking fundamentals, Linux operating system, web technologies, vulnerability assessment, penetration testing, and security tools. Practical learning through labs, Capture the Flag, and real-world testing is essential.